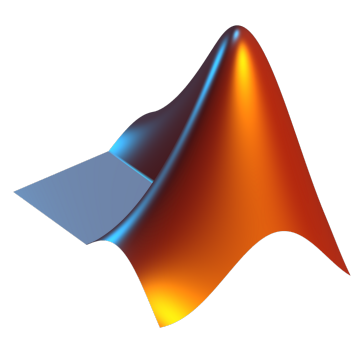




FOMC NLP: Sentiment Scores for Swap Rate Predictions



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Introduction

The objective of this study is to investigate the potential of natural language processing (NLP) to derive sentiment scores from Federal Open Market Committee (FOMC) minutes and assess whether these scores can serve as predictive indicators for federal swap rate movements. More specifically:

- Utilize NLP analysis to capture thematic sentiment scores from the textual content of FOMC minutes from 2000 to 2024.
- Construct and test statistical regression models that incorporate sentiment scores alongside 2 conventional variables, which are the current swap rate and historical VIX index, to assess the enhanced predictive power of sentiment data on future swap rates.
- Offer a novel approach to forecasting swap rates based on central bank communication by analyzing the role of FOMC sentiment in influencing market expectations.

Materials

Data Sources:

- **2-year Swap Rates:** Acquired from the Federal Reserve Bank of St. Louis (2000–2009) and Bloomberg (starting from 2009).
- **FOMC Minutes Texts:** Sourced from FOMC calendar (starting from 2019) and FOMC history (2000-2018).
- **VIX Index:** Conventional predictor for near-term market expectation obtained from Yahoo Finance (2000-2024).

Analyzing Tools:

- **FinBERT:** A BERT-based NLP model for sentiment analysis tailored to financial text.
- **MATLAB:** For computational and statistical analysis.

Methods

- **FOMC Text Processing:** Clean FOMC text data to focus on relevant economic, political, and monetary discussions and produce sentiment scores via FinBERT model accordingly.
- **Data Synchronization:** Merge the Federal Reserve swap rates from both resources, applying forward filling for sentiment scores updates to match the higher frequency of swap rate data.
- **Regression Models:** Create 2 regression models that include sentiment scores as a predictor and one that excludes them for comparison.
- **Adjusted R^2 and F-Tests:** Compare 2 models using adjusted R^2 and assess the impact of sentiment on predictive accuracy using F-tests.
- **Change Frequency:** Implement the same research on both daily and weekly 2-year swap rates.

Sentiment Scores

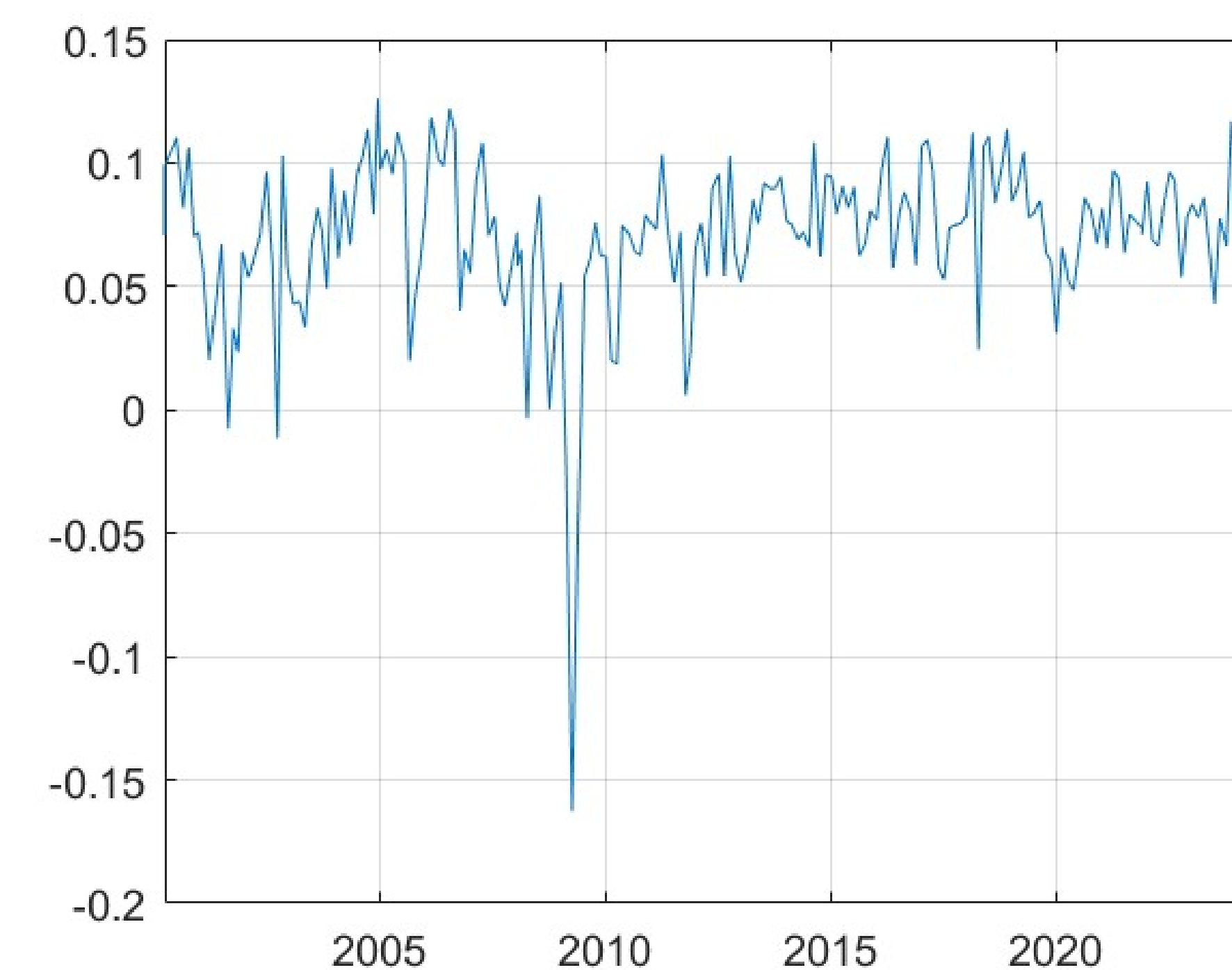


Figure 1: Sentiments Capturing

Figure 1 shows that our scoring process accurately captures the financial crisis of 2008. While it may reflect the onset of the SARS pandemic in 2002, it fails to detect sentiment regarding the COVID-19 pandemic that began in 2020, possibly due to the increasing length of the FOMC minutes text.

Model Development

Swap Rates

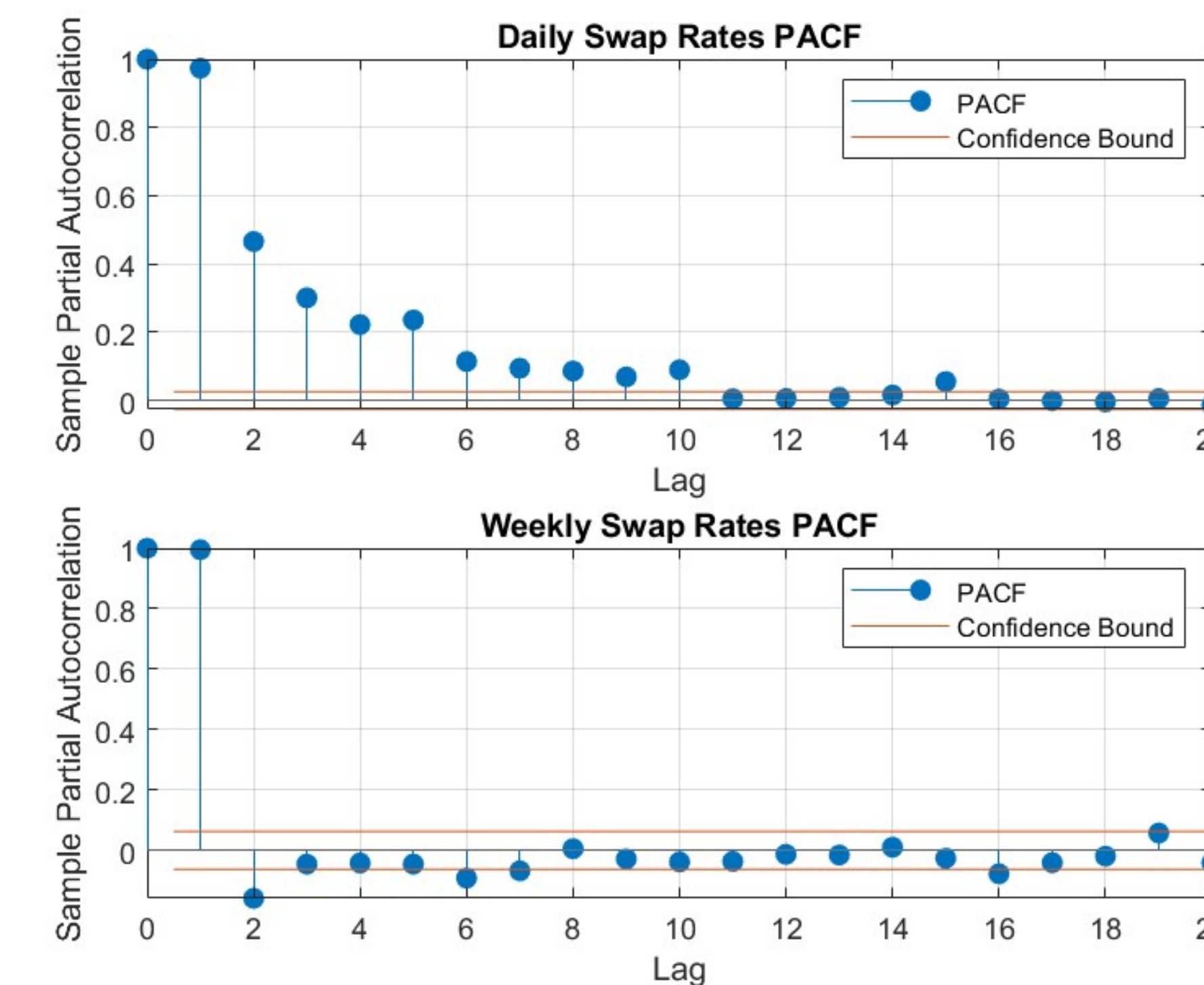


Figure 2: PACF of Swap Rates

VIX Index

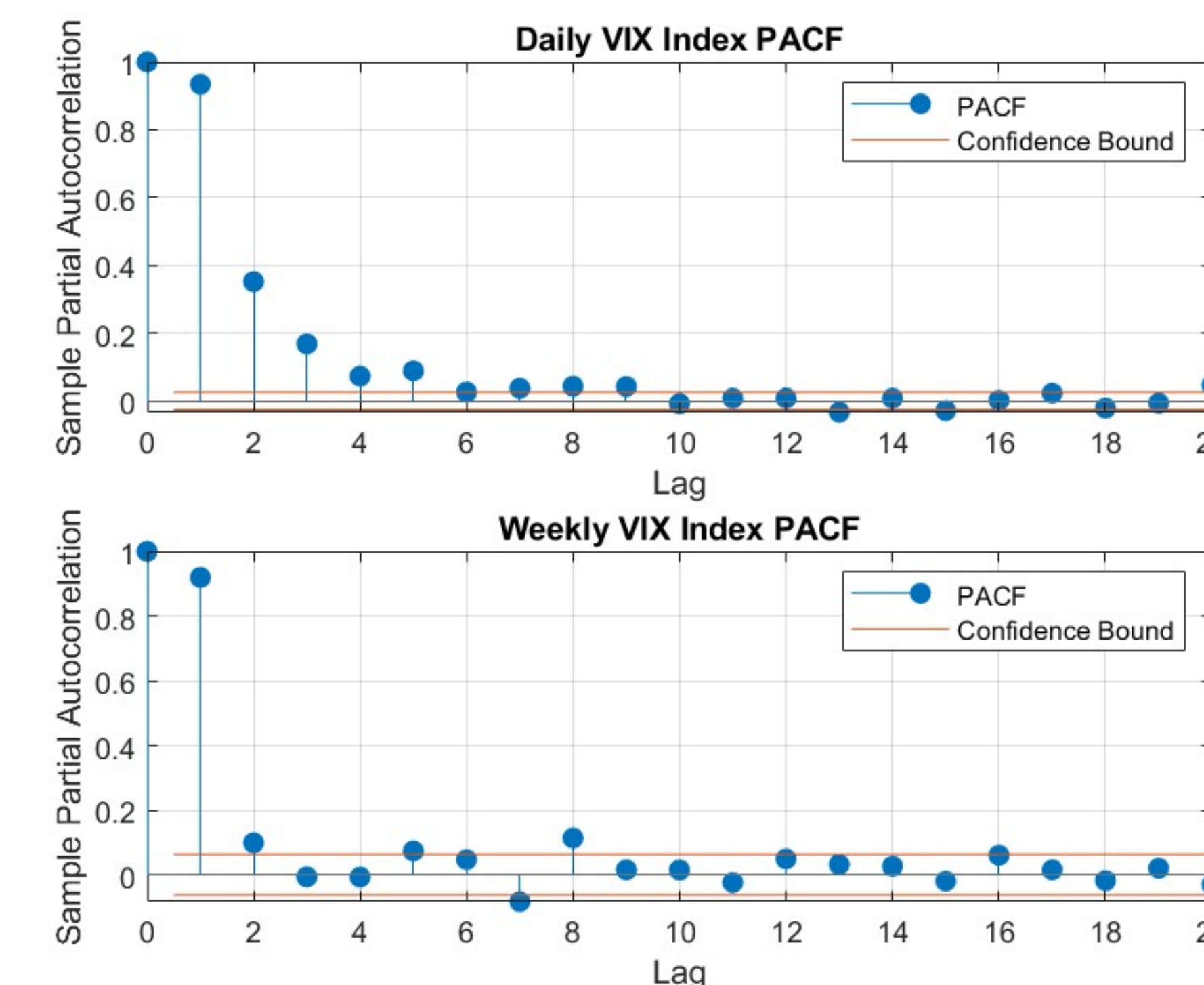


Figure 3: PACF of VIX Index

Figure 2 and 3 show that in both cases the autocorrelation is nearly 1 at a lag of 1, but quickly declines and becomes marginally insignificant at higher lags, which suggests that current swap rate and VIX index are both sufficient for the predictor from each category. The models are built accordingly:

Null:

$$\text{Swap}_{t+\Delta} \sim \text{Swap}_t + \text{VIX}_t$$

Test:

$$\text{Swap}_{t+\Delta} \sim \text{Sentiment}_t + \text{Swap}_t + \text{VIX}_t$$

Results

Daily

Variables	Estimate	SE	pValue
Swap	0.98	0.00	0.00
VIX	0.15	0.03	0.00

Table 1: Null Model

Variables	Estimate	SE	pValue
Sentiment	0.53	0.11	0.00
Swap	0.98	0.00	0.00
VIX	0.06	0.04	0.1

Table 2: Test Model

Weekly

Variables	Estimate	SE	pValue
Swap	1.00	0.00	0.00
VIX	-0.07	0.02	0.00

Table 3: Null Model

Variables	Estimate	SE	pValue
Sentiment	0.36	0.07	0.00
Swap	0.10	0.00	0.00
VIX	-0.12	0.03	0.00

Table 4: Test Model

Comparison

Models	Adj R^2	F-Stat	pValue
Daily Null	94.66%	22.48	0.00
Daily Test	94.68%		
Weekly Null	95.59%	22.63	0.00
Weekly Test	95.60%		

Conclusion

The inclusion of sentiment scores in prediction models improves accuracy. The results support the hypothesis that central bank communications contain valuable predictive information for swap rate forecasting.